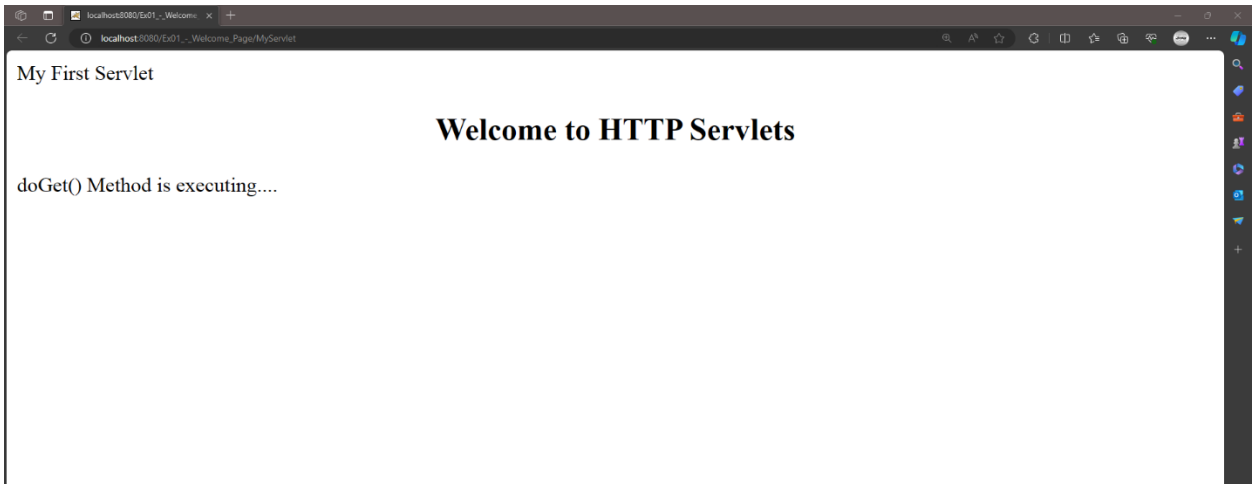
Source Code:

```
package servlet;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.io.*;
@WebServlet("/WebApplication1")
public class MyServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;
    public MyServlet() {
        super();
    }
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.print("<html><head>My First Servlet</head><body>");
        out.print("<center><h2>");
        out.print("Welcome to HTTP Servlets</h2></center>");
        out.print("<p>doGet() Method is executing....</p>");
        out.print("</body></html>");
    }
}
```

Output:


localhost:8080/Ex01_-_Welcome
localhost:8080/Ex01_-_Welcome_Page/MyServlet
My First Servlet
Welcome to HTTP Servlets
doGet() Method is executing....

Source Code:

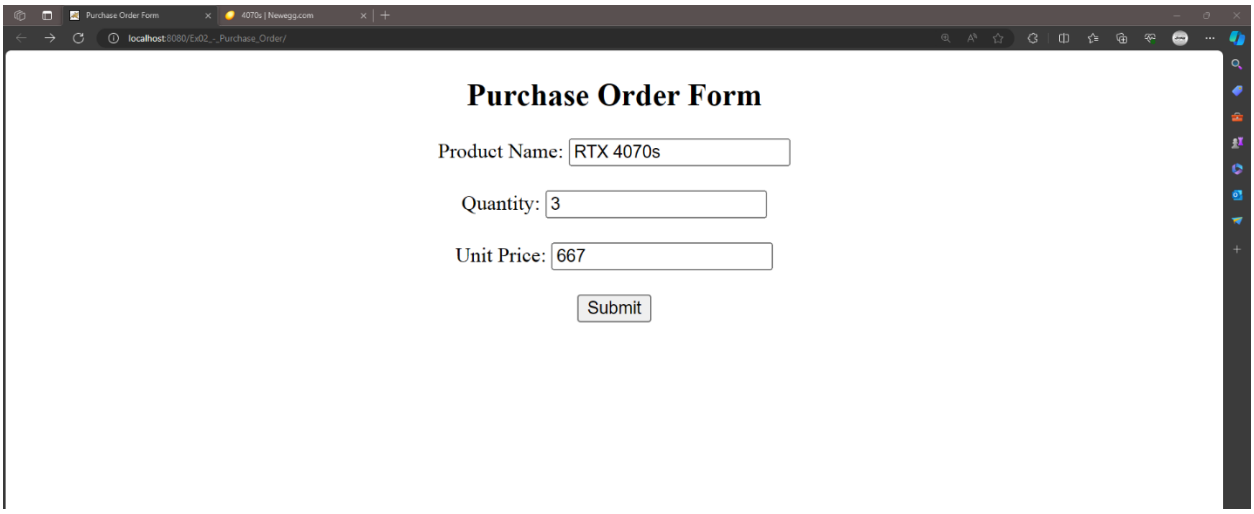
1) HTML Form (index.html):

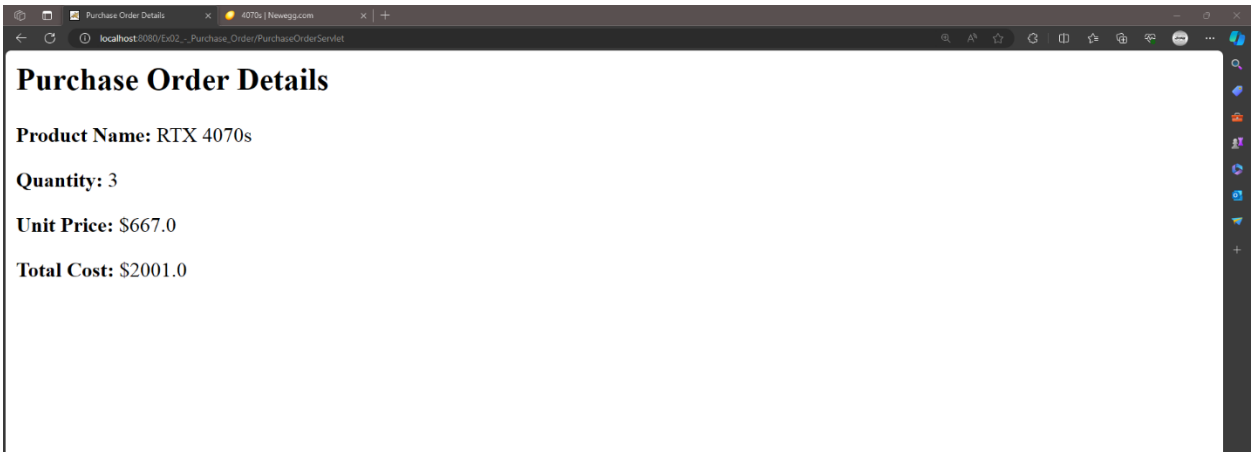
```
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta http-equiv="X-UA-Compatible" content="IE=edge">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Purchase Order Form</title>
    </head>
    <body>
        <h2>Purchase Order Form</h2>
        <form action="PurchaseOrderServlet" method="post">
            <label for="productName">Product Name:</label>
            <input type="text" id="productName" name="productName" required><br><br>
            <label for="quantity">Quantity:</label>
            <input type="number" id="quantity" name="quantity" required><br><br>
            <label for="unitPrice">Unit Price:</label>
            <input type="number" id="unitPrice" name="unitPrice" step="0.01" required><br><br>
            <input type="submit" value="Submit">
        </form>
    </body>
</html>
```

2) JAVA-Servlet(PurchaseOrderServlet.java):

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/PurchaseOrderServlet")
public class PurchaseOrderServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws IOException {
        // Retrieve form data
        String productName = request.getParameter("productName");
        int quantity = Integer.parseInt(request.getParameter("quantity"));
        double unitPrice = Double.parseDouble(request.getParameter("unitPrice"));
        // Calculate total cost
        double totalCost = quantity * unitPrice;
        // Display purchase order details
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html><head><title>Purchase Order Details</title></head><body>");
        out.println("<h2>Purchase Order Details</h2>");
        out.println("<p><strong>Product Name:</strong> " + productName + "</p>");
        out.println("<p><strong>Quantity:</strong> " + quantity + "</p>");
        out.println("<p><strong>Unit Price:</strong> $" + unitPrice + "</p>");
        out.println("<p><strong>Total Cost:</strong> $" + totalCost + "</p>");
        out.println("</body></html>");
    }
}
```

Output:


Purchase Order Form
4070s | Newegg.com
localhost:8080/Ex02_-_Purchase_Order/
Purchase Order Form
Product Name: RTX 4070s
Quantity: 3
Unit Price: 667
Submit


Purchase Order Details
4070s | Newegg.com
localhost:8080/Ex02_-_Purchase_Order/PurchaseOrderServlet
Purchase Order Details
Product Name: RTX 4070s
Quantity: 3
Unit Price: $667.0
Total Cost: $2001.0

Source Code:

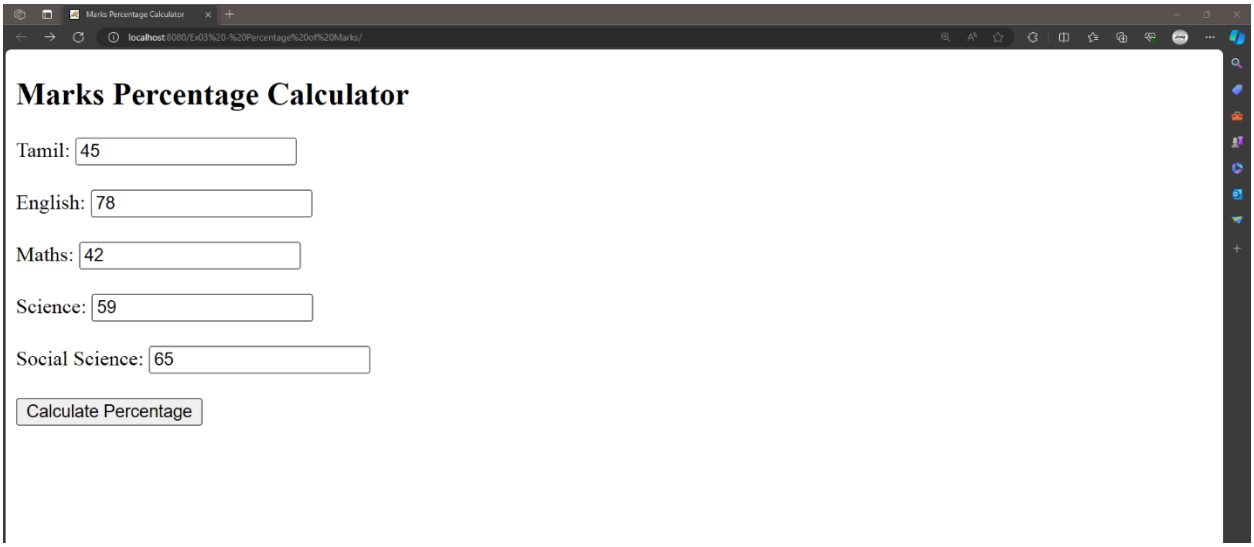
1) HTML Form (index.html):

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Marks Percentage Calculator</title>
  </head>
  <body>
    <h2>Marks Percentage Calculator</h2>
    <form action="calculatePercentage.jsp" method="post">
      <label for="tamil">Tamil:</label>
      <input type="number" id="tamil" name="tamil" required><br><br>
      <label for="english">English:</label>
      <input type="number" id="english" name="english" required><br><br>
      <label for="maths">Maths:</label>
      <input type="number" id="maths" name="maths" required><br><br>
      <label for="science">Science:</label>
      <input type="number" id="science" name="science" required><br><br>
      <label for="social_science">Social Science:</label>
      <input type="number" id="social_science" name="social_science" required><br><br>
      <input type="submit" value="Calculate Percentage">
    </form>
  </body>
</html>
```

2) JSP File (calculatePercentage.jsp):

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<% int tamil = Integer.parseInt(request.getParameter("tamil"));
   int english = Integer.parseInt(request.getParameter("english"));
   int maths = Integer.parseInt(request.getParameter("maths"));
   int science = Integer.parseInt(request.getParameter("science"));
   int social_science = Integer.parseInt(request.getParameter("social_science"));
   double totalMarks = tamil + english + maths + science + social_science;
   double percentage = (totalMarks / 500) * 100;%>
<!DOCTYPE html>
   <head>
      <title>Marks Percentage Result</title>
   </head>
   <body>
      <h2>Marks Percentage Result</h2>
      <p> <strong>Tamil:</strong> <%= tamil%></p>
      <p> <strong>English:</strong> <%= english%></p>
      <p> <strong>Maths:</strong> <%= maths%></p>
      <p> <strong>Science:</strong> <%= science%></p>
      <p> <strong>Social Science:</strong> <%= social_science%></p>
      <p> <strong>Total Marks:</strong> <%= totalMarks%></p>
      <p> <strong>Percentage:</strong> <%= percentage%> % </p>
   </body>
</html>
```

Output:


Marks Percentage Calculator
localhost:8080/Ex03%20-%20Percentage%20of%20Marks/
Marks Percentage Calculator
Tamil: 45
English: 78
Maths: 42
Science: 59
Social Science: 65
Calculate Percentage

Marks Percentage Result
localhost:8080/Ex03%20-%20Percentage%20of%20Marks/calculatePercentage.jsp
Marks Percentage Result
Tamil: 45
English: 78
Maths: 42
Science: 59
Social Science: 65
Total Marks: 289.0
Percentage: 57.8 %

Source Code:

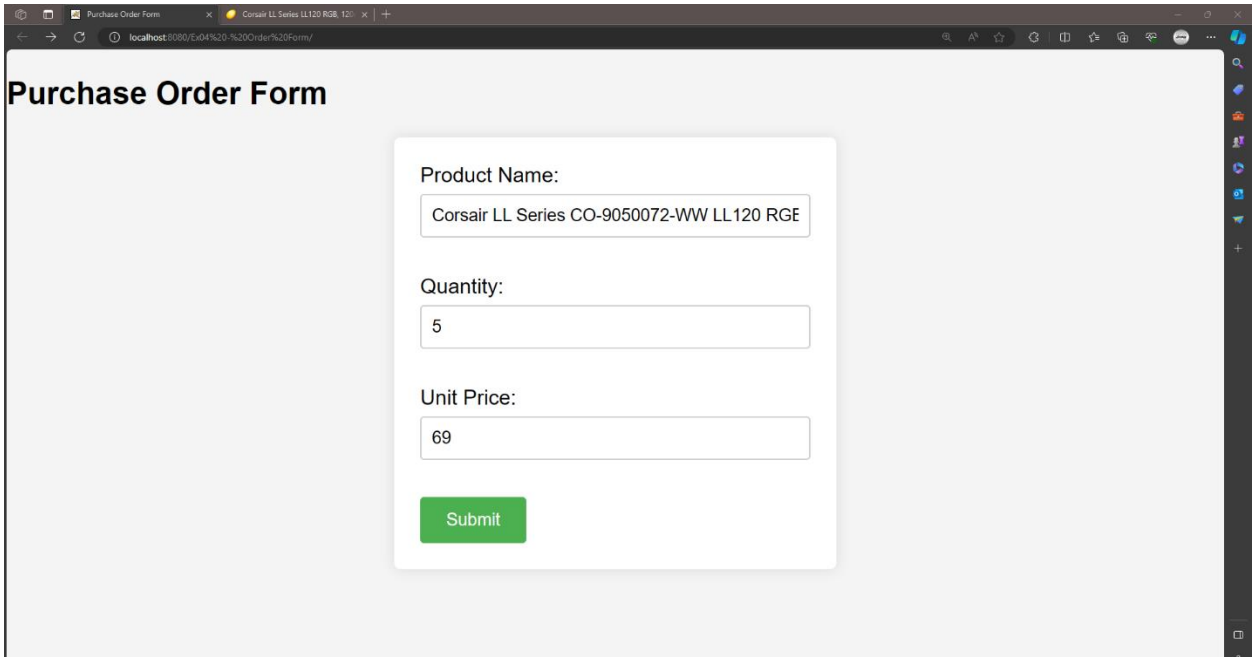
1) HTML File (index.html):

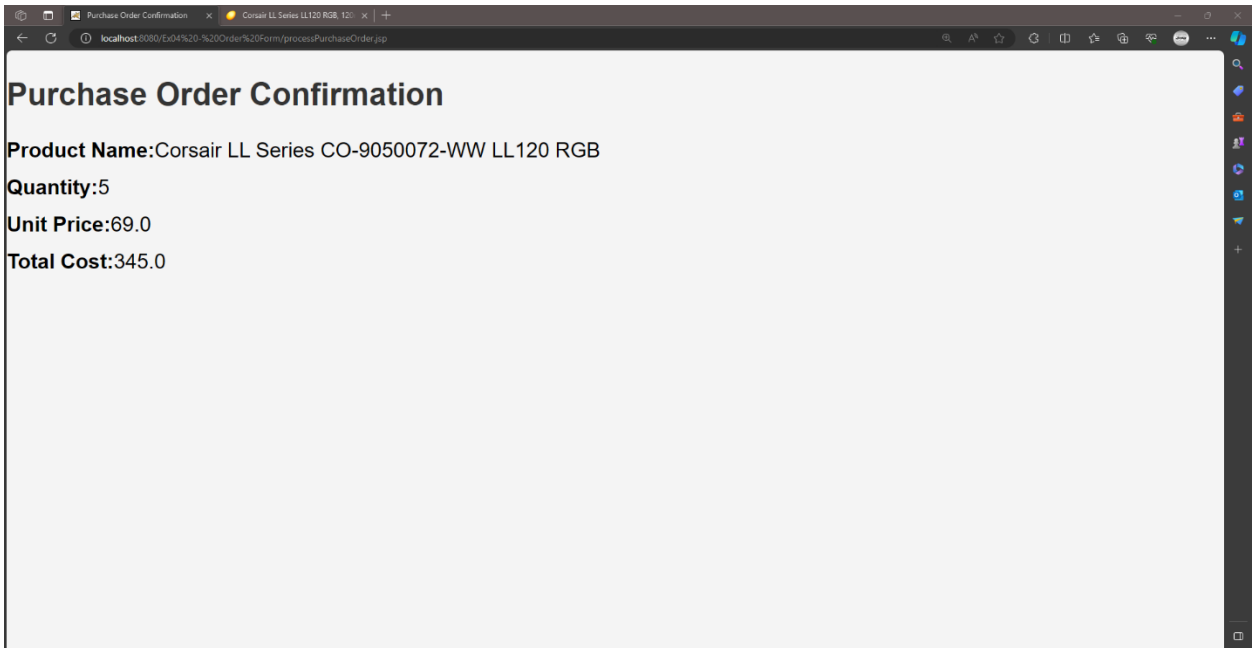
```
<!DOCTYPE html>
  <head>
    <title>Purchase Order Form</title>
  </head>
  <body>
    <h2>Purchase Order Form</h2>
    <form action="processPurchaseOrder.jsp" method="post">
      <label for="productName">Product Name:</label>
      <input type="text" id="productName" name="productName" required><br><br>
      <label for="quantity">Quantity:</label>
      <input type="number" id="quantity" name="quantity" required><br><br>
      <label for="unitPrice">Unit Price:</label>
      <input type="number" id="unitPrice" name="unitPrice" step="0.01" required><br><br>
      <input type="submit" value="Submit">
    </form>
  </body>
</html>
```

2) JSP File (processPurchaseOrder.jsp):

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="java.util.*" %>
<%  String productName = request.getParameter("productName");
   int quantity = Integer.parseInt(request.getParameter("quantity"));
   double unitPrice = Double.parseDouble(request.getParameter("unitPrice"));
   double totalCost = quantity * unitPrice;%>
<!DOCTYPE html>
   <head>
      <title>Purchase Order Confirmation</title>
   </head>
   <body>
      <h2>Purchase Order Confirmation</h2>
      <p><strong>Product Name:</strong><%= productName%></p>
      <p><strong>Quantity:</strong><%= quantity%></p>
      <p><strong>Unit Price:</strong><%= unitPrice%></p>
      <p><strong>Total Cost:</strong><%= totalCost%></p>
   </body>
</html>
```

Output:


Purchase Order Form
localhost:8080/Ex04%20-%20Order%20Form/
Purchase Order Form
Product Name:
Corsair LL Series CO-9050072-WW LL120 RGE
Quantity:
5
Unit Price:
69
Submit


Purchase Order Confirmation
localhost:8080/Ex04%20-%20Order%20Form/processPurchaseOrder.jsp
Purchase Order Confirmation
Product Name:Corsair LL Series CO-9050072-WW LL120 RGB
Quantity:5
Unit Price:69.0
Total Cost:345.0

Source Code:

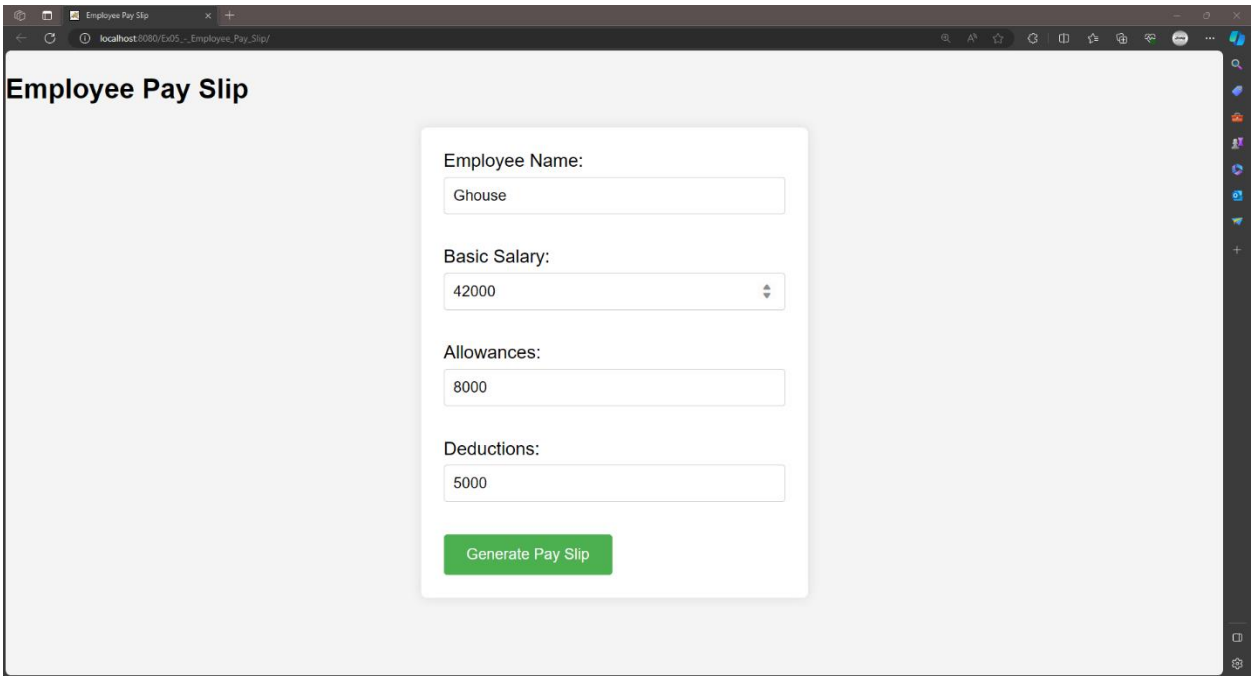
1) HTML File (index.html):

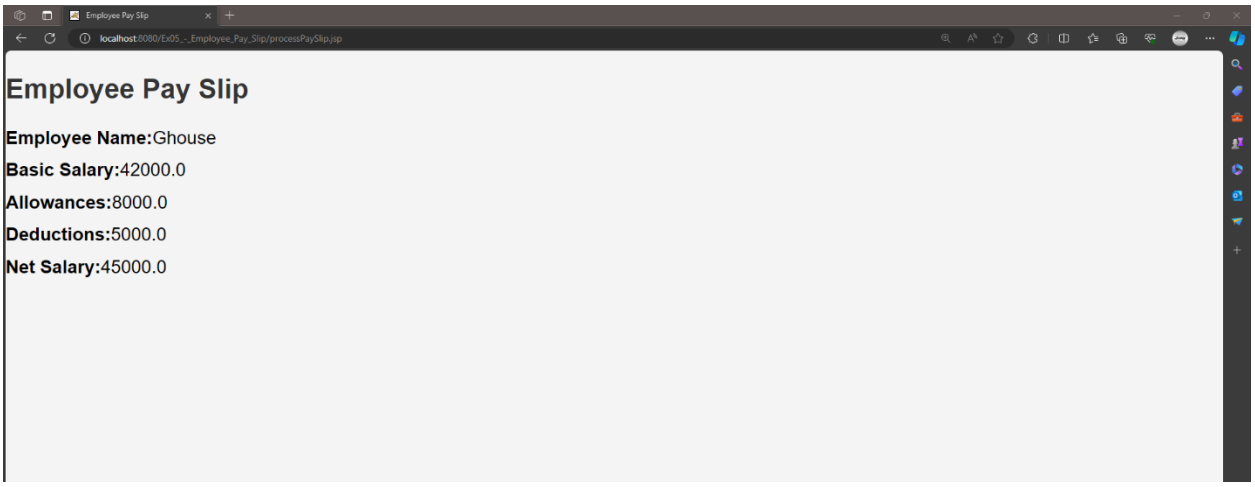
```
<!DOCTYPE html>
  <head>
    <title>Employee Pay Slip</title>
  </head>
  <body>
    <h2>Employee Pay Slip</h2>
    <form action="processPaySlip.jsp" method="post">
      <label for="employeeName">Employee Name:</label>
      <input type="text" id="employeeName" name="employeeName" required><br><br>
      <label for="basicSalary">Basic Salary:</label>
      <input type="number" id="basicSalary" name="basicSalary" required><br><br>
      <label for="allowances">Allowances:</label>
      <input type="number" id="allowances" name="allowances" required><br><br>
      <label for="deductions">Deductions:</label>
      <input type="number" id="deductions" name="deductions" required><br><br>
      <input type="submit" value="Generate Pay Slip">
    </form>
  </body>
</html>
```

2) JSP file (processPaySlip.jsp):

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%
    String employeeName = request.getParameter("employeeName");
    double basicSalary = Double.parseDouble(request.getParameter("basicSalary"));
    double allowances = Double.parseDouble(request.getParameter("allowances"));
    double deductions = Double.parseDouble(request.getParameter("deductions"));
    double netSalary = basicSalary + allowances - deductions;
%>
<!DOCTYPE html>
    <head>
        <title>Employee Pay Slip</title>
    </head>
    <body>
        <h2>Employee Pay Slip</h2>
        <p><strong>Employee Name:</strong><%= employeeName%></p>
        <p><strong>Basic Salary:</strong><%= basicSalary%></p>
        <p><strong>Allowances:</strong><%= allowances%></p>
        <p><strong>Deductions:</strong><%= deductions%></p>
        <p><strong>Net Salary:</strong><%= netSalary%></p>
    </body>
</html>
```

Output:


Employee Pay Slip
localhost:8080/Ex05_-_Employee_Pay_Slip/
Employee Pay Slip
Employee Name:
Ghouse
Basic Salary:
42000
Allowances:
8000
Deductions:
5000
Generate Pay Slip


Employee Pay Slip
localhost:8080/Ex05_-_Employee_Pay_Slip/processPaySlip.jsp
Employee Pay Slip
Employee Name:Ghouse
Basic Salary:42000.0
Allowances:8000.0
Deductions:5000.0
Net Salary:45000.0

Source Code:

1) Servlet (FormDataHandlerServlet.java):

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "FormDataHandlerServlet", urlPatterns = {"/FormDataHandlerServlet"})
public class FormDataHandlerServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        try (PrintWriter out = response.getWriter()) {
            String name = request.getParameter("name");
            String email = request.getParameter("email");
            String message = request.getParameter("message");
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Form Data</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Form Data</h1>");
            out.println("<p>Name: " + name + "</p>");
            out.println("<p>Email: " + email + "</p>");
            out.println("<p>Message: " + message + "</p>");
            out.println("</body>");
            out.println("</html>");
        }
    }
}
```

2) HTML File (index.html):

```
<!DOCTYPE html>
<html>
    <head>
        <title>Form Submission</title>
    </head>
    <body>
        <h1>Submit Form</h1>
        <form action="FormDataHandlerServlet" method="post">
            <label for="name">Name:</label><br>
            <input type="text" id="name" name="name"><br>
            <label for="email">Email:</label><br>
            <input type="email" id="email" name="email"><br>
            <label for="message">Message:</label><br>
            <textarea id="message" name="message"></textarea><br><br>
            <input type="submit" value="Submit">
        </form>
    </body>
</html>
```

Output:

Submit Form
Name:
Ghouse
Email:
Message:
Submit

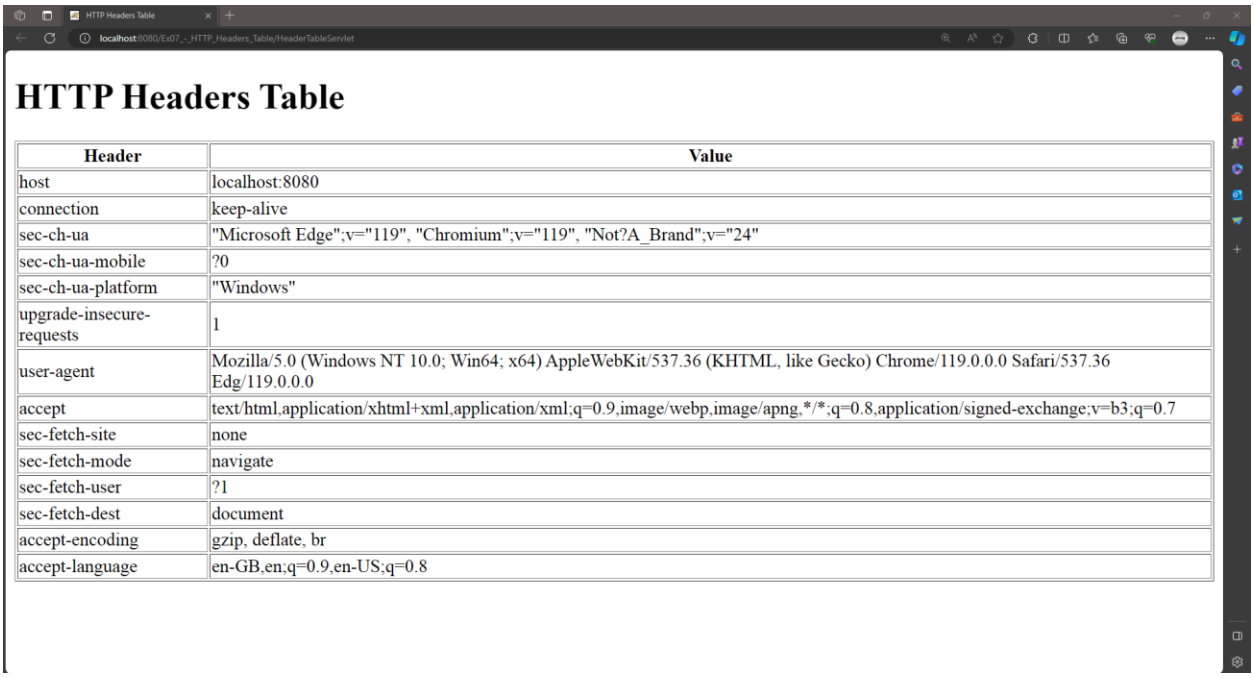
Form Data
Name: Ghouse
Email:
Message: It's enough Varsha, save some reels for tomorrow.

Source Code:

```
import java.io.IOException;
import java.io.PrintWriter;
import java.util.Collections;
import java.util.Enumeration;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "HeaderTableServlet", urlPatterns = {"/HeaderTableServlet"})
public class HeaderTableServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>HTTP Headers Table</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>HTTP Headers Table</h1>");
            out.println("<table border='1'>");
            out.println("<tr><th>Header</th><th>Value</th></tr>");
            Enumeration<String> headerNames = request.getHeaderNames();
            while (headerNames.hasMoreElements()) {
                String headerName = headerNames.nextElement();
                Enumeration<String> headerValues = request.getHeaders(headerName);
                while (headerValues.hasMoreElements()) {
                    String headerValue = headerValues.nextElement();
                    out.println("<tr><td>" + headerName + "</td><td>" + headerValue + "</td></tr>");
```

```
            }
        }
        out.println("</table>");
        out.println("</body>");
        out.println("</html>");
    }
  }
}
```

Output:



HTTP Headers Table

Header	Value
host	localhost:8080
connection	keep-alive
sec-ch-ua	"Microsoft Edge";v="119", "Chromium";v="119", "Not?A_Brand";v="24"
sec-ch-ua-mobile	?0
sec-ch-ua-platform	"Windows"
upgrade-insecure-requests	1
user-agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36 Edg/119.0.0.0
accept	text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
sec-fetch-site	none
sec-fetch-mode	navigate
sec-fetch-user	?1
sec-fetch-dest	document
accept-encoding	gzip, deflate, br
accept-language	en-GB,en;q=0.9,en-US;q=0.8

Source Code:

1) Remote Interface: (CollegeInterface.java):

```
import java.rmi.Remote;
import java.rmi.RemoteException;
public interface CollegeInterface extends Remote {
   String getStudentInfo(int studentId) throws RemoteException;
}
```

2) Server Class: (CollegeServer.java):

```
import java.rmi.RemoteException;
import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.rmi.server.UnicastRemoteObject;
import java.util.HashMap;
import java.util.Map;
public class CollegeServer extends UnicastRemoteObject implements
      CollegeInterface {
   private Map<Integer, String> studentRecords;
   public CollegeServer() throws RemoteException {
      super();
      studentRecords = new HashMap<>();
      studentRecords.put(5, "Eshwari");
      studentRecords.put(8, "Kalaiarasi");
      studentRecords.put(20, "Savitha");
   }
   public String getStudentInfo(int studentId) throws RemoteException {
      return studentRecords.getOrDefault(studentId, "Student not found");
   }
   public static void main(String[] args) {
      try {
         CollegeServer obj = new CollegeServer();
         Registry registry = LocateRegistry.createRegistry(1099);
```

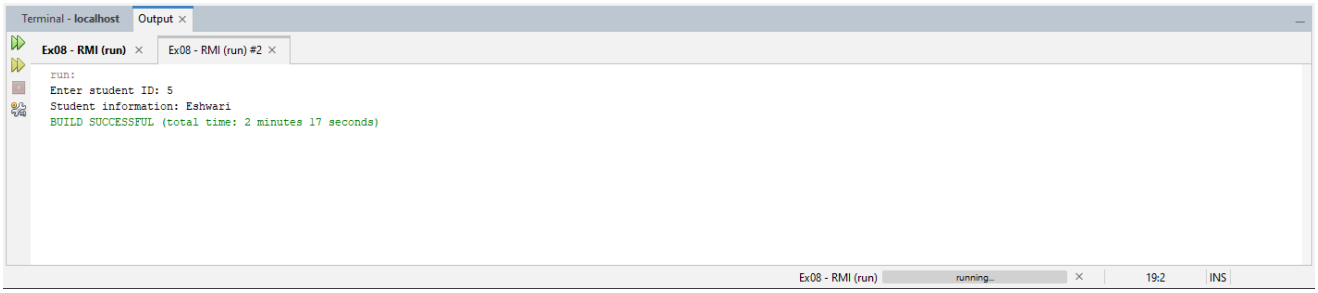
```
            registry.rebind("CollegeServer", obj);
            System.out.println("Server ready");
        } catch (Exception e) {
            System.err.println("Server exception: " + e.toString());
            e.printStackTrace();
        }
    }
}
```

3) Client Class: (CollegeClient.java)):

```
import java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.util.Scanner;
public class CollegeClient {
    public static void main(String[] args) {
        try {
            Registry registry = LocateRegistry.getRegistry("localhost", 1099);
            CollegeInterface stub = (CollegeInterface) registry.lookup("CollegeServer");
            Scanner scanner = new Scanner(System.in);
            System.out.print("Enter student ID: ");
            int studentId = scanner.nextInt();
            String studentInfo = stub.getStudentInfo(studentId);
            System.out.println("Student information: " + studentInfo);
        } catch (Exception e) {
            System.err.println("Client exception: " + e.toString());
            e.printStackTrace();
        }
    }
}
```

Output:

Terminal - localhost
Output - Ex08 - RMI (run)
run:
Server ready
Ex08 - RMI (run)
running...
12:42
INS


Terminal - localhost
Output
Ex08 - RMI (run)
Ex08 - RMI (run) #2
run:
Enter student ID: 5
Student information: Eshwari
BUILD SUCCESSFUL (total time: 2 minutes 17 seconds)
Ex08 - RMI (run)
running...
19:2
INS

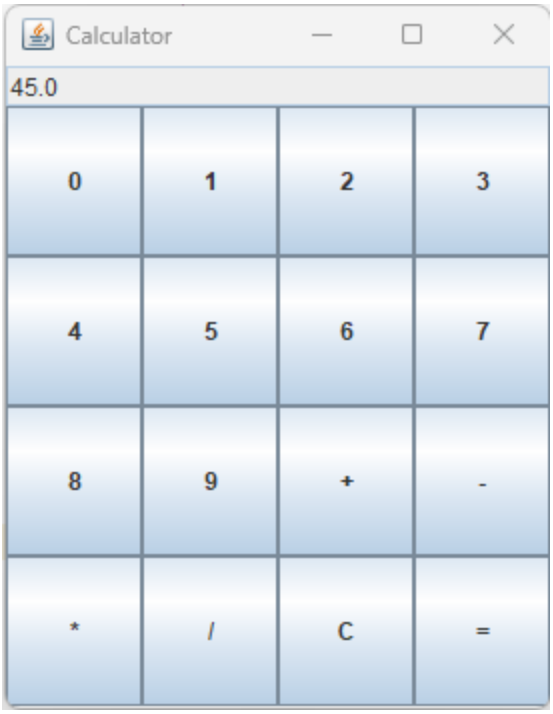
Source Code:

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
public class Calculator extends JApplet implements ActionListener {
    private JTextField textField;
    private JButton[] numberButtons;
    private JButton[] operatorButtons;
    private JButton clearButton;
    private JButton equalsButton;
    private String displayText = "";
    private double result = 0;
    private String operator = "";
    public void init() {
        setLayout(new BorderLayout());
        textField = new JTextField();
        textField.setEditable(false);
        add(textField, BorderLayout.NORTH);
        JPanel buttonPanel = new JPanel();
        buttonPanel.setLayout(new GridLayout(4, 4));
        numberButtons = new JButton[10];
        for (int i = 0; i < 10; i++) {
            numberButtons[i] = new JButton(String.valueOf(i));
            numberButtons[i].addActionListener(this);
            buttonPanel.add(numberButtons[i]);
        }
        operatorButtons = new JButton[4];
        operatorButtons[0] = new JButton("+");
        operatorButtons[1] = new JButton("-");
        operatorButtons[2] = new JButton("*");
        operatorButtons[3] = new JButton("/");
        for (int i = 0; i < 4; i++) {
```

```
        operatorButtons[i].addActionListener(this);
        buttonPanel.add(operatorButtons[i]);
    }
    clearButton = new JButton("C");
    clearButton.addActionListener(this);
    buttonPanel.add(clearButton);
    equalsButton = new JButton("=");
    equalsButton.addActionListener(this);
    buttonPanel.add(equalsButton);
    add(buttonPanel, BorderLayout.CENTER);
}
public void actionPerformed(ActionEvent e) {
    JButton source = (JButton) e.getSource();
    String buttonText = source.getText();
    if (buttonText.equals("C")) {
        displayText = "";
        result = 0;
        operator = "";
    } else if (buttonText.equals("=")) {
        double secondOperand = Double.parseDouble(displayText);
        switch (operator) {
            case "+":
                result += secondOperand;
                break;
            case "-":
                result -= secondOperand;
                break;
            case "*":
                result *= secondOperand;
                break;
            case "/":
                if (secondOperand != 0) {
                    result /= secondOperand;
```

```
                } else {
                    displayText = "Error: Division by zero";
                }
                break;
            }
            displayText = String.valueOf(result);
            operator = "";
        } else if (buttonText.equals("+") || buttonText.equals("-")
                || buttonText.equals("*") || buttonText.equals("/")) {
            operator = buttonText;
            result = Double.parseDouble(displayText);
            displayText = "";
        } else {
            displayText += buttonText;
        }
        textField.setText(displayText);
    }
}
```

Output:



Source Code:

1) Receiver Project (Main.java):

```
import java.io.DataOutputStream;
import java.net.Socket;
public class Main {
    public static void main(String[] args) {
        try {
            String messageToSend = "Que Ota? (Translation: Want another one?)";
            Socket socket = new Socket("127.0.0.1", 12345);
            DataOutputStream out = new DataOutputStream(socket.getOutputStream());
            out.writeUTF(messageToSend);
            out.flush();
            out.close();
            socket.close();
            System.out.println("Message sent successfully.");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

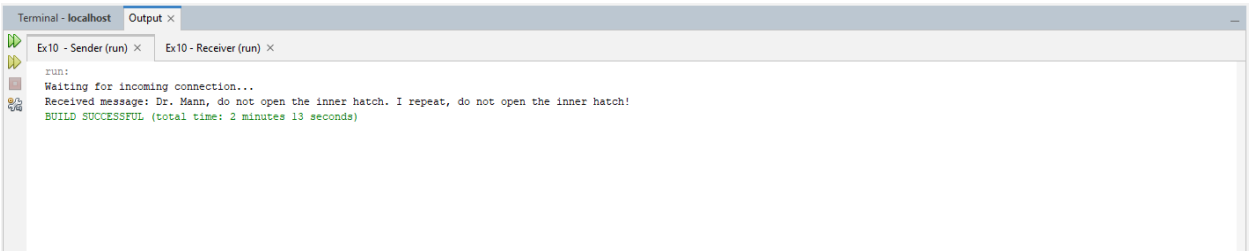
2) Sender Project (Main.java):

```
import java.io.DataInputStream;
import java.net.ServerSocket;
import java.net.Socket;
public class Main {
    public static void main(String[] args) {
        try {
            ServerSocket serverSocket = new ServerSocket(12345);
            System.out.println("Waiting for incoming connection...");
            Socket socket = serverSocket.accept();
            DataInputStream in = new DataInputStream(socket.getInputStream());
            String receivedMessage = in.readUTF();
            System.out.println("Received message: " + receivedMessage);
            in.close();
            socket.close();
            serverSocket.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output:

Terminal - localhost
Output - Ex10 - Sender (run)
run:
Waiting for incoming connection...
Ex10 - Sender (run)
17:33
INS

Terminal - localhost
Output
Ex10 - Sender (run)
Ex10 - Receiver (run)
run:
Message sent successfully.
BUILD SUCCESSFUL (total time: 0 seconds)
6:114
INS


Terminal - localhost
Output
Ex10 - Sender (run)
Ex10 - Receiver (run)
run:
Waiting for incoming connection...
Received message: Dr. Mann, do not open the inner hatch. I repeat, do not open the inner hatch!
BUILD SUCCESSFUL (total time: 2 minutes 13 seconds)